

ARK 12S CAN ESC

Single channel electronic speed controller with CAN FD

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NDAA compliant, made in the USA.

PRELIMINARY. This product is still under validation. All specifications are subject to change without notice and some values are provisional or marked TBD. Confirm anything critical with ARK Electronics before committing to a design.

A single motor, three phase speed controller that reports telemetry over CAN FD and also accepts a conventional single wire signal input, so it can operate as a CAN node or as a drop in replacement for a PWM/DShot ESC. All high current connections are made at threaded SMT standoffs, so the ESC bolts directly to a bus bar or motor mount.

Features

- 4S – 12S battery input, six 120V 1.7mΩ MOSFETs
- CAN FD to 5Mbps, software switchable 120Ω termination, dual connectors for daisy chaining
- 0 – 300A current and 0 – 102V battery voltage measurement
- Sensorless commutation from three phase dividers and a virtual neutral node
- 380μF of 100V ceramic bulk capacitance — no electrolytics to derate with age
- Hardware overcurrent protection with fault classification on the debug console

Specifications

Parameter	Value
Motor outputs	1 motor, 3 phases
Battery input	4S – 12S lithium polymer. 9V minimum, 50.4V nominal maximum, 70V absolute maximum
Continuous / burst current	TBD — see Current Ratings
Current measurement	0 – 300A. 100μΩ low side shunt, sense amplifier at 100V/V, 10mV/A
Voltage measurement	0 – 102V. 30kΩ / 1kΩ divider, 31V/V, 3.3V clamp
Phase sensing	20kΩ / 1kΩ per phase with 20kΩ virtual neutral, 21V/V, 3.3V clamp
Bulk capacitance	380μF, 100V ceramic
Operating temperature	-40°C to +85°C
Dimensions / weight	57.75mm × 36.20mm × 9.04mm, 22g
Construction	8 layers, 1.6mm finished, 2oz outer / 1oz inner copper, filled via in pad, ENIG, black mask with white silkscreen

The board operates down to 9V, but 4S remains the minimum recommended cell count — below that there is very little margin between a sagging pack under load and the low voltage cutoff.

Current Ratings

Continuous and burst current ratings are TBD. These are thermal limits, not silicon limits — the current this ESC holds indefinitely depends on how it is mounted, cooled, and heatsinked. Ratings will be published once thermal validation is complete. Until then treat the overcurrent trip below as a fault threshold, not an operating target.

Overcurrent is detected by monitoring MOSFET drain to source voltage against a 700mV threshold. That voltage develops across a resistance that rises with temperature, so the trip current falls as the power stage heats up:

Junction temperature	25°C	100°C	150°C	175°C
Trip current	412A	292A	233A	206A

Firmware applies its own limits well below the hardware trip: **200A** current limit, with temperature foldback from **105°C** over a **15°C** band. Foldback reduces power progressively rather than cutting output abruptly.

Board Maximum Design Specifications

Hardware limits the board is built to. These are not operating recommendations — the continuous rating will land well below the overcurrent trip.

Parameter	Design limit	Set by
MOSFET drain to source voltage	120V	Power MOSFETs
Gate driver supply voltage	102V	Gate driver
Bulk capacitor voltage	100V	Input bulk capacitors
Buck regulator input voltage	120V	Primary step down regulator
Input TVS standoff voltage	70V	Input transient suppressor, 77.8V breakdown
Voltage measurement range	102V	31V/V divider
Current measurement range	300A	Shunt and current sense amplifier
Signal input	5.5V	5V tolerant, ESD protected

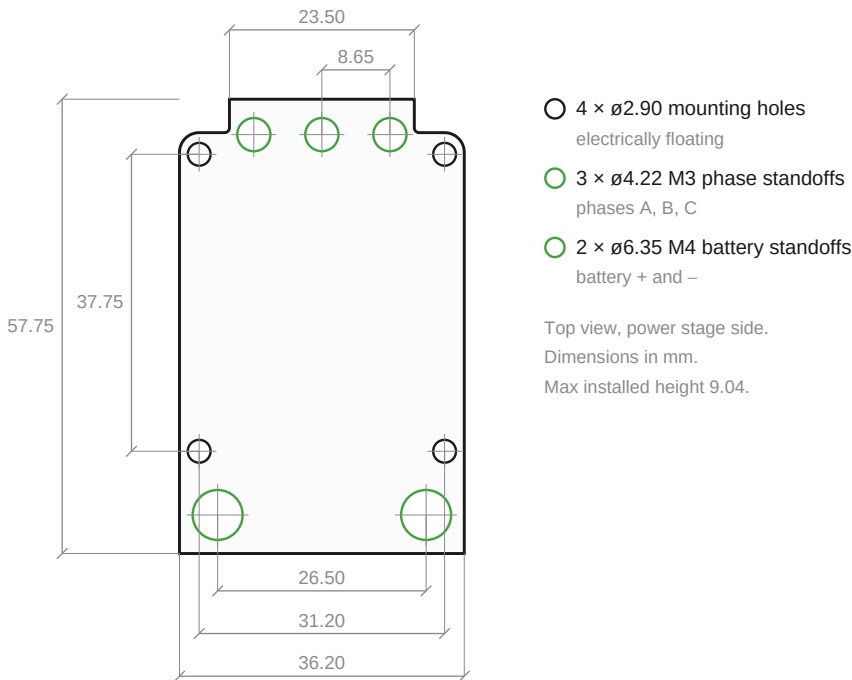
The 70V suppressor threshold and the 102V gate driver limit sit close together. A large input transient clamped by the suppressor can still exceed the gate driver rating, which is why external bulk capacitance matters on long battery leads. On 12S, or with leads longer than a few inches, add a bulk capacitor at the battery input.

Connectors

Reference	Connector	Function
J1, J4	4 pin JST GH, 1.25mm	CAN FD, wired in parallel for daisy chaining
J2	5 pin JST SR, 1.0mm	Flight controller: signal input, telemetry, analog current, battery voltage
J3	6 pin JST SR, 1.0mm	SWD and debug console, 115200 baud transmit only

J2 carries unregulated battery voltage on pin 1, not 5V. On 12S that is over 50V, unfused and present whenever the battery is connected. Many ESCs put 5V on this pin — confirm what your flight controller expects before making up a cable. Pin assignments for all three connectors are published at docs.arkelectron.com.

Mechanical



Board outline and hole pattern, generated from the released STEP model. Hole positions are to centre.

Feature	Detail
Mounting	4 × $\varnothing 2.90$ mm for M2 hardware on a 31.20mm × 37.75mm pattern. Electrically floating — metal hardware creates no ground loop and carries no ESC current
Battery terminals	M1 positive, M2 negative — M4 threaded standoffs on 26.50mm centers
Phase terminals	M3 phase C, M4 phase A, M5 phase B — M3 threaded standoffs on 8.65mm centers

Making power connections

The standoffs are threaded and act as captive nuts, so they sit on the opposite side of the board from the side the screw enters. Either put a ring or fork terminal on the pad and drive a screw through it into the standoff, capturing the PCB between terminal and standoff, or solder the wire directly to the pad on the side opposite the standoff. The standoffs are round and cannot be counter-held with a wrench — they do not need to be, since they are soldered during SMT and the screw clamps the board rather than twisting them.

Battery negative reaches board ground through the 100 $\mu\Omega$ current shunt, so the M2 standoff and board ground are **not** the same node. Do not bond them together externally and do not use a chassis or frame path as a return for battery current — either shorts out the shunt and the ESC loses its current measurement.

Firmware is loaded over SWD, over CAN with the DroneCAN GUI Tool, automatically by a PX4 flight controller from its SD card, or with the ARK32 Configurator at [ark32.arkelectron.com](https://docs.arkelectron.com). Full procedures, pinouts, and console reference are at docs.arkelectron.com.

Preliminary document. All specifications, ratings, and mechanical dimensions are subject to change without notice. Continuous and burst current ratings are pending thermal validation and are not published here.